

## UNIT 3

### **Cyber Offences and Countermeasures**

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**Part-A:** Introduction to cyber offences and cybercrime, how criminals plan the attacks, social engineering, cyber stalking, cyber cafe and cybercrimes, Botnets: The fuel for cybercrime, attack vector. introduction to cloud security.

## Course Outcomes (COs)

Upon completion of course the students will be able to

- **analyze cyber offences** and implement effective countermeasures

# Introduction

- The rapid growth of computers, smartphones, and the Internet has transformed communication, business, education, and government services.
- This digital transformation has also led to an increase in illegal activities conducted through cyberspace. These illegal activities are known as **cyber offences** and **cybercrime**.
- They pose serious threats to individuals, organizations, and governments by compromising confidentiality, integrity, and availability of information.

# Cybercrime

- **Definition:**

Cybercrime is any criminal activity in which a computer, network, or digital device is used as a **tool**, **target**, or **both** to commit an offence.

- **Examples:**

- Hacking into a company's database
- Online banking fraud
- Identity theft
- Ransomware attacks
- Phishing emails
- Spreading computer viruses

# Cyber Offence

- A **cyber offence** is any act that violates cyber laws and is committed using computers, communication networks, or electronic devices. Every cyber offence is punishable under applicable cyber laws.
- **Examples of cyber offences:**
  - Unauthorized access to computer systems
  - Data theft
  - Email spoofing
  - Cyberstalking
  - Website defacement
  - Distribution of malicious software

# **Classification of Cybercrime**

- 1. Crimes Against Individuals**
- 2. Crimes Against Property**
- 3. Crimes Against Organizations**
- 4. Crimes Against Government**

# Common Types of Cyber Offences

| Cyber Offence      | Description                                                            |
|--------------------|------------------------------------------------------------------------|
| Hacking            | Gaining unauthorized access to a computer or network.                  |
| Phishing           | Sending fake emails or websites to steal personal information.         |
| Identity Theft     | Misusing another person's personal information.                        |
| Malware            | Creating or spreading malicious software.                              |
| Ransomware         | Encrypting files and demanding payment for their release.              |
| Cyberstalking      | Repeated online harassment or intimidation.                            |
| Email Spoofing     | Sending emails using a fake sender identity.                           |
| Data Theft         | Stealing confidential or sensitive information.                        |
| Website Defacement | Altering the appearance or content of a website without authorization. |

# Causes of Cybercrime

- Weak passwords
- Lack of cybersecurity awareness
- Software vulnerabilities
- Poor security policies
- Financial gain
- Revenge or personal motives
- Political or ideological reasons
- Rapid growth of Internet services



# Prevention of Cybercrime

- Use strong and unique passwords.
- Enable Multi-Factor Authentication (MFA).
- Update operating systems and software regularly.
- Install antivirus and firewall protection.
- Avoid clicking suspicious emails or links.
- Encrypt sensitive data.
- Back up important files regularly.
- Conduct cybersecurity awareness training.
- Monitor systems for unusual activities.

# How Criminals plan the attack?

- Criminals use many methods and tools to identify vulnerabilities of their target.
- Criminals plan attacks were categorised into
  - Active
  - Passive
  - Inside
  - Outside
- **Active Attacks:**
  - Def: To alter the system
  - Affects: Availability, Integrity and authenticity of data.
- **Passive Attacks:**
  - Def: To gain information about the target.
  - Affects: Confidentiality
- **Inside Attack :**
  - Def: Attempted within the security perimeter of an organisation.
  - Affects: Access more resources than expected.
- **Outside attack:**
  - Def: Attempted by a source outside the security perimeter (insider/outsider) who is indirectly associated with the organisation.
  - Sources: Either from internet or a remote access connection.

## **Phases to plan cyber crime**

- Information gathering. (passive attack)
- Reconnaissance : To gain information about an enemy .
- Starts with Foot printing.
- Scanning and scrutinizing the information.
  - For the validity of the information.
  - To identify the existing vulnerabilities.
- Launching an attack
  - Gaining and maintaining the system access.

## Passive Attack

- Information gathering about a target without their knowledge.
  - Ex: Watching a building to identify employees entering time
- It is done using internet search or googling to gain individual or company information.
- Google or Yahoo search to locate information about employees.
  - Tools:
    - **GOOGLE EARTH:** *It is a virtual map of globe and geographic information program. It maps the earth by the superimposition of images obtained from satellite imagery and provides aerial photography of the globe*  
Ex: ([www.earth.google.com](http://www.earth.google.com)).
    - **INTERNET ARCHIVE:** *The internet archive is an internet library with the purpose of offering permanent access for researchers, historians and scholars to historical collections that exist in digital format (Ramot et al., 2003). It includes texts, Audio moving images and software as well as archived web pages in our collections.*  
Ex: ([www.archive.org](http://www.archive.org)).
    - **PEOPLE SEARCH:** *People search provides details about personal information, date of birth, residential address, contact number etc.* Ex: ([www.whitepagesinc.com](http://www.whitepagesinc.com)).
    - **DOMAIN NAME CONFIRMATION:** *To perform searches for domain names using multiple keywords, this helps to find every registered domain name in “com”, “net”, “org”, “edu” ([www.namedroppers.com](http://www.namedroppers.com)).*
    - **WHOIS:** *This is a domain registration lookup tool. This utility is used for communicating with WHOIS servers located around the world to obtain domain registration information. WHOIS supports IP address queries and automatically selects the appropriate server for IP address. This tool will lookup information on a domain IP address or a domain registration server or you can use the default option which will select a server for you ([www.whois.net](http://www.whois.net)).*
    - **NSLOOKUP:** *The name nslookup means “name server lookup”, the tool is used on windows and Unix to query details, including ip addresses of a particular computer and other technical details such as mail exchanger records for a domain and name servers of a domain ([www.nslookup.downloadsoftware4free.com](http://www.nslookup.downloadsoftware4free.com)).*
    - **eMailTrackerPro:** *eMailTrackerPro analyzes the E-mail header and provides the IP address of the system that send the mail ([www.emailtrackerpro.com](http://www.emailtrackerpro.com)).*
    - **HTTrack:** *This tool acts like an offline browser. It can mirror the entire website to a desktop. One can analyze the entire websites by being offline ([www.httrack.com](http://www.httrack.com)).*

## Active Attacks

- **ARPING:** This is a network tool that
  - Broadcasts arp packets and receive replies similar to “ping”.
  - It is good for mapping a local network and finding used IP space
  - It broadcasts, who has an arp packet on the network and prints answers.
  - It is very useful when trying to pick an unused IP for a net to which routing does not exist as yet
- **Bing:** This tool is used
  - for pinging the bandwidth.
  - It is one of the point-to-point bandwidth measurement tool
  - It can calculate the blank throughput between any couple of networks. (<http://ai3.asti.dost.gov.ph/sat/bing.html>).
- It is also called “Rattling the doorknobs” or “Active reconnaissance”
- It can provide confirmation to an attacker about security measures in place .

## Tools:

- **Dsniff :** This is network auditing tool to capture username, password and authentication information on a local subnet.
- **DNStracer:** This tool is used
  - for determining the data source of a DNS server
  - Notifies the continuous process of DNS server
- **Hmap:** This tool is used
  - To get the fingerprinting of web servers
    - to identify the version of server, vender of the server, model of the server and more
- **Nmap:** This tool is used
  - to scan the port address of the network,
  - It takes the fingerprint of operating system version, service and it can scan the network rapidly (<http://insecure.org/nmap>).

## Scanning and scrutinizing gathered Information

- **Objectives of scanning:**
  - Port Scanning : Where information goes in and out of a computer
  - Identify open/close ports and services
  - Network Scanning: Understand IP addresses and computer network systems information.
  - Vulnerability Scanning: Understand the existing weaknesses in the system.
- Scrutinizing : “Enumeration”
  - Objectives:
  - Valid user or group
  - Network resources and /or shared resources
  - OS and different applications that are running on the OS

| S.No | Port Number | Port Description             |
|------|-------------|------------------------------|
| 1    | 1           | TCP Port service Multiplexer |
| 2    | 18          | Message send protocol        |
| 3    | 20          | FTP data                     |
| 4    | 25          | SMTP                         |
| 5    | 43          | WHOIS                        |

## Launch the Attack

- After scan and enumeration, the attack is launched using the following steps:
  - Crack the password
  - Exploit the privileges
  - Execute the malicious commands/applications
  - Hide the files
  - Cover the tracks like delete the access logs.
- Note:
  - The attackers spend 90% of the time in Reconnaissance, Scanning and scrutinizing on a target.
  - Only 10% of the time in launching the attack.

# Social Engineering

- It is a way for criminals to gain access to information systems.
- The purpose of social engineering is usually to secretly install spyware, other malicious software or to trick persons into handing over passwords and/or other sensitive financial or personal information

## **What are they looking for**

- Obtaining simple information such as your pet's name, where you're from, the places you've visited; information that you'd give out freely to your friends.
- Take a close look at some of the 'secure' sites you log into. Some have a 'secret question' you have to answer, if you cannot remember your username or password. The questions seem pretty tough for an outsider looking into trying to hack into your account.
  - What's the name of your first pet?
  - What is your maiden name?
  - When was your mother/father born?
  - Where were you born?

*Do these sound familiar?*



# Social Engineering

- It is the “technique to influence” and “persuasion to deceive” people to obtain the information or perform some action.
- Social engineers exploit the natural tendency of a person to trust social engineers’ word, rather than exploiting computer security holes.
- Social engineering involves gaining sensitive information or unauthorized access privileges by building inappropriate trust relationships with insiders.
- The sign of truly successful social engineers is that they receive information without any suspicion.

## **Classification of Social Engineering**

### ***1. Human-Based Social Engineering***

Human-based social engineering refers to person-to-person interaction to get the required/desired information.

### ***2. Computer-Based Social Engineering***

Computer-based social engineering refers to an attempt made to get the required/desired information by using computer software/Internet.

## 1. Human Based (Non Technical)

1. Impersonating an employee or valid user : Posing employee of same organisation.
2. Posing as an important user : Pretend to be an important user
3. Using a third person : By authorised source to use system
4. Calling technical support : By using Help desk
5. Shoulder surfing : By watching over a person shoulder
6. Dumpster diving : Looking in trash for information



## **2. Computer Based Social Engineering**

1. Fake EMails
2. Email Attachments
3. Pop-up Windows

## **Protecting Yourself from Social Engineering Attack**

1. Network defenses to repel virus
  - Virus protection (McAfee, Norton, Symantec, etc...)
  - Email attachment scanning
  - Firewalls, etc...
2. Organizations must decide what information is sensitive
3. Security must be periodically tested
4. Contact your security office immediately if you have any concerns at work

# Cyber Stalking

## Definition::

It is a crime in which the attacker harasses a victim using electronic communication, such as e-mail or instant messaging (IM), or messages posted to a Web site or a discussion group.

## Stalking :

- A cyberstalker relies upon the anonymity afforded by the Internet to allow them to **stalk** their victim without being detected.



- Cyberstalking messages differ from ordinary spam.
- Cyberstalker targets a specific victim with often threatening messages, while the spammer targets a multitude of recipients with simply annoying messages.

## **Types of Stalkers**

- Mainly 2 types :
  - Online Stalkers :
    - Def: Interaction with the victim using Internet.
    - Ex: Email, Chat
  - Offline Stalkers:
    - Def: Begin the attack using traditional methods.
    - Ex: Watching daily, Personal websites.

## **Other types :**

- The four most important types are:
  - Rejected Stalker
  - Resentful Stalker
  - Predatory Stalker
  - Intimacy Stalker

## **Rejected Stalker**

- Most common, persistent, and intrusive
- Obsessed with someone who is a former romantic partner or friend, and who has ended their relationship, or indicates that he or she intends to end the relationship.

## **Resentful Stalker**

- Looking for revenge against someone who has upset them--it could be someone known to the stalker or a complete stranger.
  - Their behaviors are meant to frighten and distress the victim

## **Predatory Stalker**

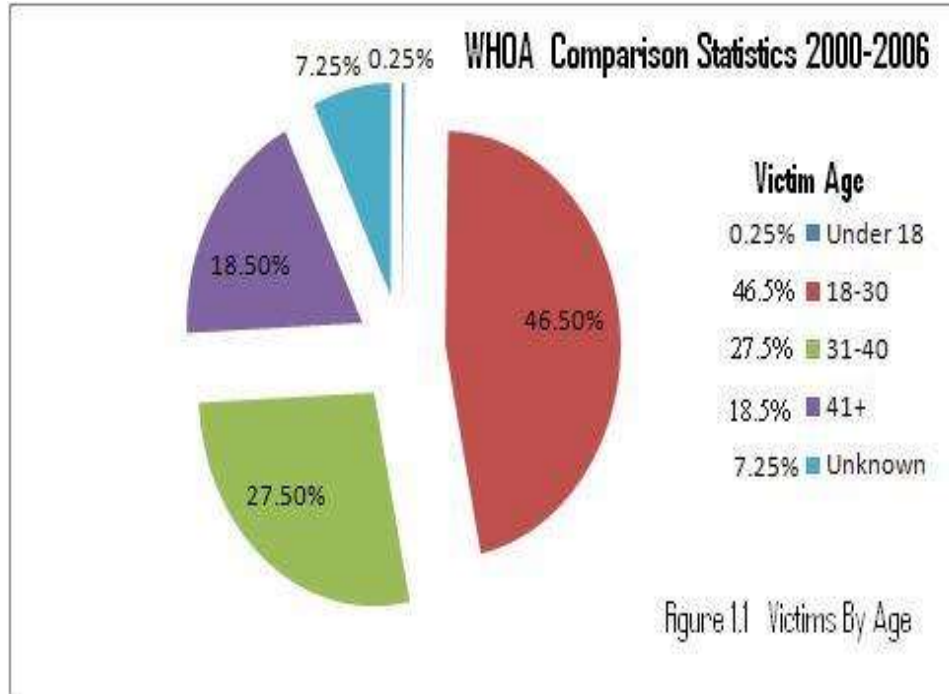
- Least common
- Are a classic sexual predator whose plan is to physically or sexually attack the victim.
- Their motivated purely by the desire for sexual gratification and power over their victim

## **Intimacy Stalker**

- They seeks to establish an intimate, loving relationship with their victim.
  - The victim and himself were “meant to be together.”
- These types of people think that the victim owes them love and affection because of all the time and effort it took for the stalker to stalk them.

## Analysis...

- The majority of cyberstalkers are men and the majority of their victims are women but.....
- Jane A. Hitchcock, president of WHOA says, “The most surprising thing we've seen is the rise in female cyberstalkers - this increased from 27% in 2000 to 35% in 2002 to 38% in 2003.”
- According to the Working to Halt Online Abuse (WHOA) from 2000 to 2006, out of the total cases, 2036, of cyberstalking....





**STALKED  
BEFORE AGE 25**



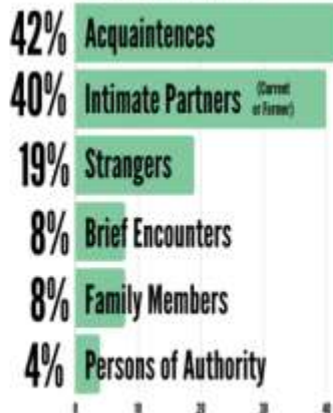
**STALKED  
BEFORE AGE 18**

DVSN.org

Source: SPARC



### STALKERS



**1 IN 3 STALKERS  
HAVE STALKED BEFORE**



## How stalking works?

- In the following way
  1. Personal information gathering about the victim.
    - Ex: Name, background, contact number, etc.
  2. Establish a contact with victim through phone.
  3. Stalkers will almost always establish a contact with the victims through EMail.
  4. Some stalkers keep on sending repeated E-mail.
  5. Post victims personal information on any illicit services website
  6. Who were received information , start calling the victims on the given contact details.
  7. Stalkers subscribe/register the E-mail account of the victim to pornographic websites.
- 1. Real Life Incident:
  - Case study: In Delhi, 40 calls in 3 days.

## Cybercafe and Cybercrimes

- Cybercrimes such as stealing of bank passwords and subsequent fraudulent withdrawal of money have also happened through cybercafes.
- Cybercafes have also been used regularly for sending obscene mails to harass people.
- Indian Information Technology Act (ITA) 2000 interprets cybercafes as “network service providers” referred to under the erstwhile Section 79, which imposed on them a responsibility for “due diligence” failing which they would be liable for the offenses committed in their network.


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### Secure Banking

- ▶ Security Measures
- ▶ Browser Requirements
- ▶ Our Unique Features
- ▶ Secure Your PC
- ▶ Do's & Don'ts

### Secure Yourself While..

- ▶ Using Mobile Banking
- ▶ Using your ATM/Debit Card
- ▶ Using your Credit Card
- ▶ Using Internet Banking
- ▶ Shopping Online

### Learn More

- ▶ Types of Fraud
- ▶ Identify Fraud
- ▶ Cyber Cafe Security
- ▶ Password-Related Tips
- ▶ Privacy Policy
- ▶ FAQs
- ▶ Glossary

### Contact US

- ▶ Do-not-Call
- ▶ Contact Points



128-bit Secure Socket Layer Encryption Technology  
Entrust Digital Certificate  
Secured Bill Pay & Funds Transfer  
Tips for Customers

## Cyber Cafe Security

If you are accessing any website (including ICICIBANK.com) from cyber cafe, any shared computer or from a computer other than that of your own, please change your passwords after such use from your own PC at workplace or at home.

It is very important to do so especially when you have entered your transaction password from such shared computer or cybercafe computer. Change these Passwords from your own PC at workplace or at house.

### Login

- ▶ Personal
- ▶ Corporate
- ▶ Money2India
- ▶ Young Stars

### Forgot Password

- ▶ New user ?
- ▶ Forgot user ID & Password

### New User - Register Now

### Internet Banking Demo

### Online Security

a Savings account  
that turns into an fd  
and biab returns

**Figure 1**

Cybercafe security.

Source: <http://www.icicibank.com/pfsuser/temp/cybersec.htm> (27 June 2009).



## Virtual Keyboard for Internet Banking

At ICICI Bank, We are committed to make your banking with us a safe and wonderful experience. We provide you with Virtual Keyboard to Protect your password. Virtual Keyboard is an online application to enter password with the help of a mouse.

### Advantage of a Virtual Keyboard

The Virtual Keyboard is designed to protect your password form malicious "Spyware" and "Trojan Programs". Use of Virtual keyboard will reduce the risk of password theft.

### Process To Use Virtual Keyboard

Steps to use Virtual keyboard are as follows:

- Enter Login ID using Physical Keyboard.
- Select the check box 'Use Virtual Keyboard'.
- Use the Virtual Keyboard to the login password.
- Once you have entered your password, click "Log-in".
- 

Functions of different keys on the Virtual Keyboard

**Caps Lock:** This key can be used to enter upper case if the password consist of capital letters.

**Back Space:** This key will clear the last character entered in the password field.

**Clear:** This key wil clear all characters entered in the password field by Virtual keyboard.

**Tab:** This key is visible only for change or forced change password. field by Virtual keyboard. This key can be used to ente values in the next field.

**Figure 2**

Virtual keyboard.

Source: <http://www.icicibank.com/pfsuser/webnews/virtualkeyboad.htm> (27 June 2009).

Cybercriminals can either install malicious programs such as keyloggers and/or Spyware or launch an attack on the target.

Here are a few tips for safety and security while using the computer in a cybercafe:

1. Always logout
2. Stay with the computer
3. Clear history and temporary files
4. Be alert
5. Avoid online financial transactions
6. Change passwords
7. Virtual keyboard
8. Security warnings

# Beyond the Syllabus:

Awareness Theory



# 10 Common Tricks Used by Scammers:

- Beware and Stay Safe: Scammers target people of all ages, but middle-aged and elderly individuals are particularly vulnerable. Be cautious and aware of these common tricks:
  1. **TRAI Phone Scam:** Fraudsters claim to be from TRAI, stating your mobile number is linked to illegal activities, and services will be suspended. -  
Reality: TRAI doesn't suspend services; telecom companies do.
  2. **Parcel Stuck at Customs:** Scammers claim a parcel with contraband has been intercepted and demand payment. -  
Action: Disconnect and report the number.
  3. **Digital Arrest:** Fake police officers threaten digital arrest or online interrogation. -  
Reality: Police don't conduct digital arrests or online interrogations.

4. **Family Member Arrested:** Scammers claim a relative will be arrested and demand payment.

-  
Action: Verify with family members before taking action.

5. **Get Rich Quick Trading:** Social media ads promise high returns on stock investments. -

Reality: High-return schemes are likely scams.

6. **Easy Tasks for Big Rewards:** Scammers offer high sums for simple tasks, then ask for investment. -

Reality: Easy money schemes are scams.

7. **Credit Card Issued in Your Name:** Fake executives confirm large transactions on bogus credit cards. -

Action: Check with your bank.

8. **Mistaken Money Transfer:** Scammers claim incorrect transactions and ask for refunds.

Action: Verify transactions with your bank.

9. **KYC Expired:** Scammers ask for KYC updates via links. -

Reality: Banks require in-person KYC updates.

10. **Generous Tax Refund:** Fraudsters pose as tax officials, asking for bank details.

Reality: Tax departments already have bank details and communicate directly.



# Stay Safe:

- 1. Verify information before acting.
- 2. Don't click suspicious links.
- 3. Confirm transactions with banks.
- 4. Report suspicious calls/numbers.
- 5. Be cautious of high-return schemes.
- 6. Update KYC in-person.
- 7. Don't share personal/bank details.

# Report Scams:

- 1: GOI Chakshu portal @ <https://sancharsaathi.gov.in/sfc/>
- 2a. National Consumer Helpline (1800-11-4000)
- 2b. Cyber Crime Reporting Portal (cybercrime.gov.in)
- 3. Local Police Station

**Stay informed, stay vigilant!**



दूरसंचार विभाग  
DEPARTMENT OF TELECOMMUNICATIONS



india.gov.in



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KEEP YOURSELF AWARE

FAQs

IN SOCIAL MEDIA

IMAGE GALLERY

USEFUL LINKS



## चक्षु - Report Suspected Fraud & Unsolicited Commercial Communication

### Report Suspected Fraud Communication

(Received within last 30 days)

Chakshu facilitates citizens to report the suspected fraud communications with the intention of defrauding telecom service users for cyber-crime, financial frauds, non-bonafide purpose like impersonation or any other misuse through Call, SMS or WhatsApp.

Few examples of suspected fraud communications are communication related to Bank Account / Payment Wallet / SIM / Gas connection / Electricity

### Report Unsolicited Commercial Communication (UCC) / Spam

(Report within 3 days for action)

Chakshu facilitates citizens to report UCC or spam received through Voice Call or SMS which is not as per the consent given by recipient to sender or as per registered preference (s). UCC / Spam are dealt as per The Telecom Commercial Communication Customer Preference Regulation (TCCPR), 2018 regulations of Telecom Regulatory Authority of India (TRAI). Visit <https://traigov.in/what-spam-or-ucc> for more details.

[Continue reporting →](#)

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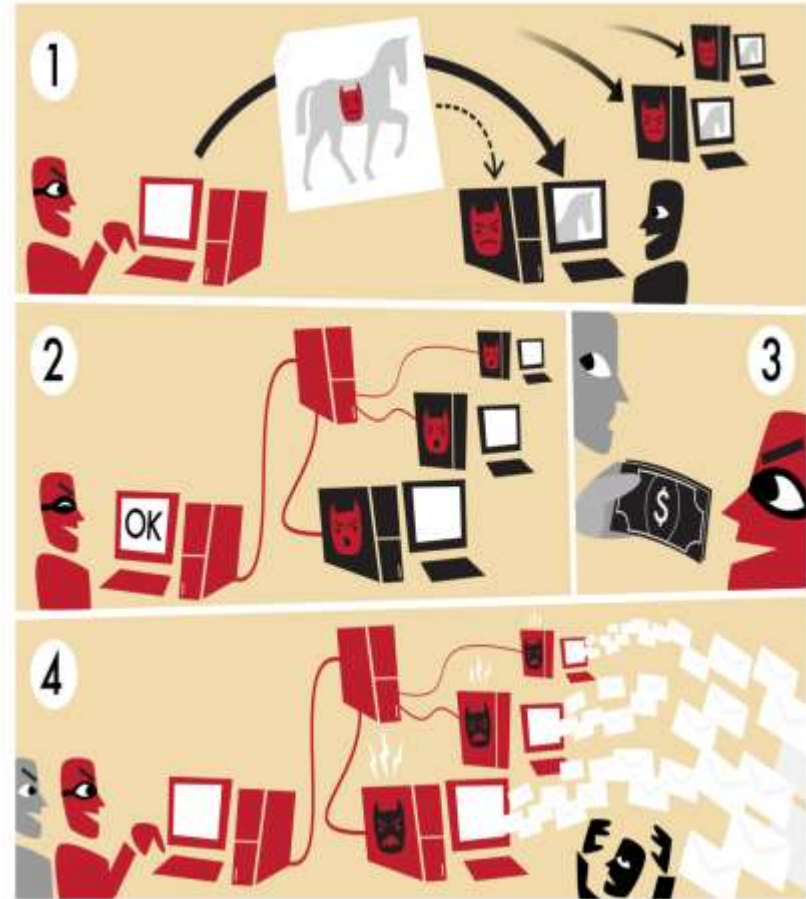
Financial Fraud  
Brochures final.pdf

# Botnet: The fuel of Cybercrime

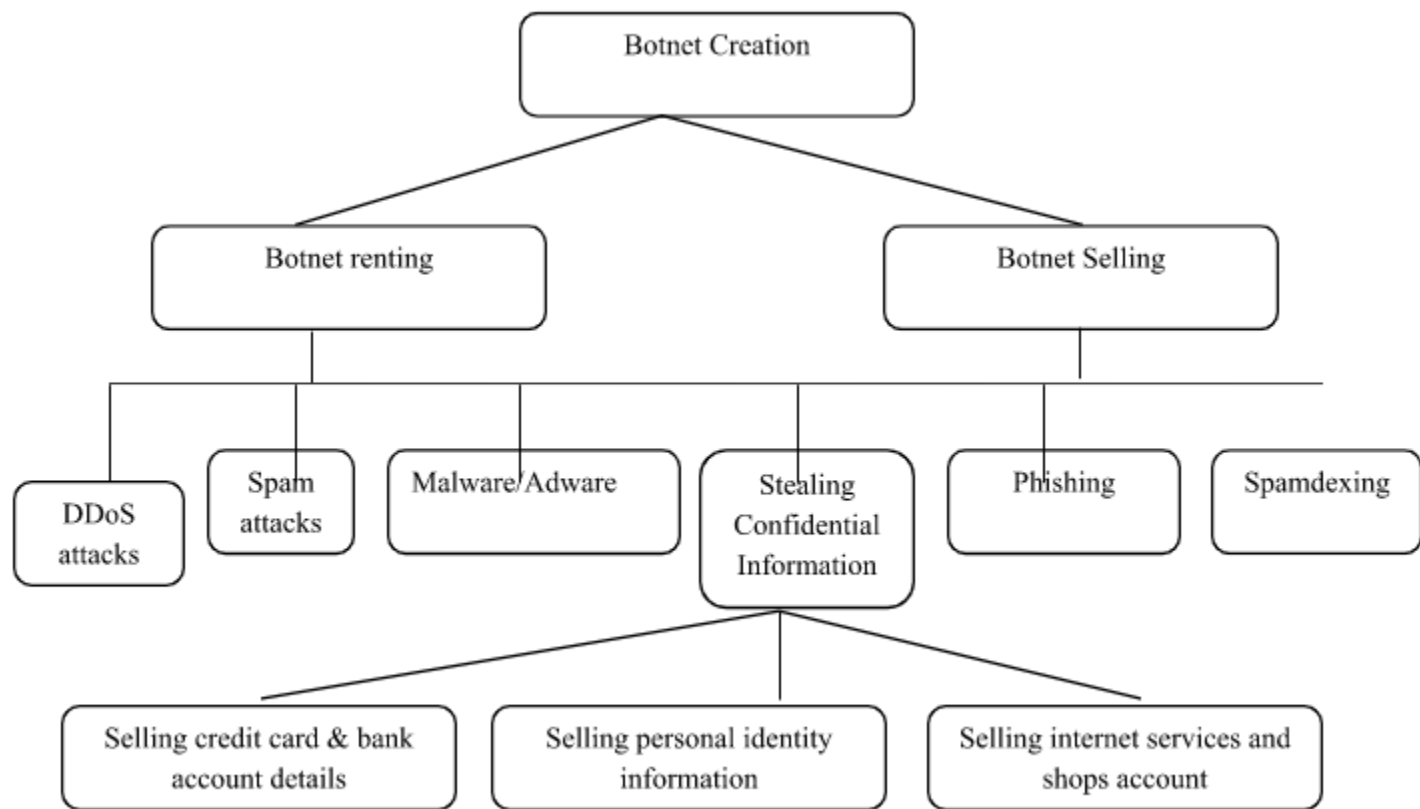
- A Botnet (also called as zombie network) is a network of computers infected with a malicious program that allows cybercriminals to control the infected machines remotely without the users' knowledge.
- Your computer system maybe a part of a Botnet even though it appears to be operating normally.
- Botnets are often used to conduct a range of activities, from distributing Spam and viruses to conducting denial-of-service (DoS) attacks.

## Evolution of Botnets

- Motivation change in computer hacking
  - Vandalism
  - Financial gains



# How botnets creates the business



One can ensure following to secure the system:

1. Use antivirus and anti-Spyware software and keep it up-to-date.
2. Set the OS to download and install security patches automatically.
3. Use a firewall to protect the system from hacking attacks while it is connected on the Internet.
4. Disconnect from the Internet when you are away from your computer.
5. Downloading the freeware only from websites that are known and trustworthy
6. Check regularly the folders in the mail box – “sent items” or “outgoing” – for those messages you did not send.
7. Take an immediate action if your system is infected.



# Attack vector

- An attack vector is the path, method, or technique that a cyber attacker uses to gain unauthorized access to a computer system, network, or application in order to steal data, spread malware, or disrupt services.

# Common Attack Vectors

| • Attack Vector            | • Description                                                                    | • Example                                 |
|----------------------------|----------------------------------------------------------------------------------|-------------------------------------------|
| • Phishing                 | • Fake emails or messages that trick users into revealing sensitive information. | • Fake bank login email                   |
| • Weak Passwords           | • Easy-to-guess passwords allow unauthorized access.                             | • Password: "123456"                      |
| • Malware                  | • Malicious software installed through downloads or attachments.                 | • Opening an infected email attachment    |
| • Software Vulnerabilities | • Exploiting bugs in outdated software.                                          | • Attacking an unpatched operating system |
| • Social Engineering       | • Manipulating people into revealing confidential information.                   | • Pretending to be IT support             |
| • Insider Threats          | • Employees intentionally or accidentally misuse access.                         | • Employee copying confidential data      |
| • USB Devices              | • Infected USB drives spread malware.                                            | • Plugging in an unknown USB drive        |
| • Misconfigured Systems    | • Incorrect security settings expose systems.                                    | • Publicly accessible database            |

# How an Attack Vector Works

- The attacker identifies a weakness.
- The attacker selects an attack vector.
- The attacker exploits the weakness.
- The attacker gains unauthorized access.
- The attacker steals data, installs malware, or disrupts services.

# Cloud Security

Cloud computing services, while offering considerable benefits and cost savings makes it easier for cybercriminals to attack these systems.

Cloud computing is Internet (“cloud”)-based development and use of computer technology (“ computing”).

A cloud service has three distinct characteristics which differentiate it from traditional hosting:

1. It is sold on demand
2. it is elastic in terms of usage
3. the service is fully managed by the provider.

## **Advantages of Cloud Computing**

Cloud computing has following advantages:

1. Applications and data can be accessed from anywhere at any time.
2. It could bring hardware costs down.
3. Organizations do not have to buy a set of software or software licenses for every employee and the organizations could pay a fee to a cloud computing company.
4. Organizations do not have to rent a physical space to store servers and databases. Servers and digital storage devices take up space.
5. Organizations would be able to save money on IT support because organizations will have to ensure about the desktop (i.e., a client) and continuous Internet connectivity instead of servers and other hardware.

The cloud computing services can be either private or public.

- A public cloud sells services to anyone on the Internet.
- A private cloud is like a proprietary network or a data center that supplies the hosted services to a limited number of people.

## **Types of Services**

Services provided by cloud computing are as follows:

1. Infrastructure-as-a-service (IaaS)
2. Platform-as-a-service (PaaS)
3. Software-as-a-service (SaaS)

**Table 1** | Cloud computing service providers

| <i>Sr. No.</i> | <i>Service Providers</i>                                                                                                                                                                                               | <i>Weblink</i>                                                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.             | Amazon: It offers flexible, simple, and easy computing environment in the cloud that allows development of applications.                                                                                               | <a href="http://aws.amazon.com/ec2/">http://aws.amazon.com/ec2/</a>                                                                                             |
| 2.             | 3Tera: It offers AppLogic grid OS that enables infrastructure solutions according to the changing needs of business.                                                                                                   | <a href="http://www.3tera.com/">http://www.3tera.com/</a>                                                                                                       |
| 3.             | Force.com: It allows building of core business applications like enterprise resource planning (ERP), human resource management (HRM), and supply chain management (SCM).                                               | <a href="http://www.salesforce.com/platform/">http://www.salesforce.com/platform/</a>                                                                           |
| 4.             | Appistry-Cloud Computing Middleware: It allows easily scalable cloud computing for a wide variety of applications and services for both public and private clouds.                                                     | <a href="http://www.appistry.com/">http://www.appistry.com/</a>                                                                                                 |
| 5.             | Microsoft Live Mesh: This cloud setup synchronizes the files with the all users' devices like laptop, Mac, mobile phone, or others and allows to access the files from any device as well as enables sharing of files. | <a href="https://www.mesh.com/Welcome/default.aspx">https://www.mesh.com/Welcome/default.aspx</a>                                                               |
| 6.             | AppNexus: This helps a user to launch several operating systems, run a variety of applications, load balance these applications, and store huge amount of secure data.                                                 | <a href="http://www.appnexus.com/">http://www.appnexus.com/</a>                                                                                                 |
| 7.             | Flexiscale: It is self-service through control panel or API – features full self-service – start/stop/delete, change memory/CPU/storage/IPs of virtual dedicated servers.                                              | <a href="http://www.flexiscale.com/">http://www.flexiscale.com/</a>                                                                                             |
| 8.             | GoogleApp Engine: This is a free setup that allows the users to run their web application on Google infrastructure.                                                                                                    | <a href="http://www.google.com/apps/intl/en/business/index.html">http://www.google.com/apps/intl/en/business/index.html</a>                                     |
| 9.             | GoGrid: It offers unique multiserver control panel that enables the user to deploy and manage load-balanced cloud servers.                                                                                             | <a href="http://www.gogrid.com/">http://www.gogrid.com/</a>                                                                                                     |
| 10.            | Terremark Enterprise Cloud: It provides the power to the user for computing resources for user's mission-critical applications.                                                                                        | <a href="http://www.terremark.com/services/cloudcomputing/theenterprisecloud.aspx">http://www.terremark.com/services/cloudcomputing/theenterprisecloud.aspx</a> |

## **Cybercrime and Cloud Computing**

Prime area of the risk in cloud computing is protection of user data.

Table 2 shows the major areas of concerns in cloud computing domain.

**Table 2** | Risks associated with cloud computing environment

| <i>Sr. No.</i> | <i>Area</i>           | <i>What is the Risk?</i>                                                                                                                                                                                                                                                                                                                                                                                         | <i>How to Remediate the Risk?</i>                                                                                                                                                                                                                                                                            |
|----------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.             | Elevated user access  | Any data processed outside the organization brings with it an inherent level of risk, as outsourced services may bypass the physical, logical, and personnel controls and will have elevated user access to such data.                                                                                                                                                                                           | Customer should obtain as much information as he/she can about the service provider who will be managing the data and scrutinizing vendor's monitoring mechanism about hiring and oversight of privileged administrators, and IT controls over the access privileges.                                        |
| 2.             | Regulatory compliance | Cloud computing service providers are not able and/or not willing to undergo external assessments. This can result into non-compliance with various standards/laws like the US government's Health Insurance Portability and Accountability Act (HIPAA), or Sarbanes-Oxley; the European Union's Data Protection Directive or the credit card industry's Payment Card Industry Data Security Standard (PCI DSS). | The organization is entirely responsible for the security and integrity of their own data, even when it is held by a service provider. Hence, organization should force cloud computing service providers to undergo external audits and/or security certifications and submit the report on periodic basis. |
| 3.             | Location of the data  | The organizations that are obtaining cloud computing services may not be aware about where the data is hosted and may not even know in which country it is hosted.                                                                                                                                                                                                                                               | Organizations should ensure that the service provider is committed to obey local privacy requirements on behalf of the organization to store and process the data in the specific jurisdictions.                                                                                                             |

*(Continued)*



*(Table 2 continued)*

|    |                                        |                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                   |
|----|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. | Segregation of data                    | As the data will be stored under stored environment, encryption mechanism should be strong enough to segregate the data from other organizations, whose data are also stored under the same server.                | Organization should be aware of the arrangements made by the service provider about segregation of the data. In case of encryption mechanism, the service provider should display encryption schemes and testing of the mechanism by the experts.                                                 |
| 5. | Recovery of the data                   | Business continuity in case of any disaster – availability of the services and data without any disruption. Application environment and IT infrastructure across multiple sites are vulnerable to a total failure. | Organization should ensure the enforcement of contractual liability over the service provider about complete restoration of data within stipulated timeframe. Organization should also be aware of Business Continuity Plan/Disaster Recovery Plan (BCP/DRP) established by the service provider. |
| 6. | Information security violation reports | Due to complex IT environment and several customers logging in and logging out of the hosts, it becomes difficult to trace inappropriate and/or illegal activity.                                                  | Organization should enforce the contractual liability toward providing security violation logs at frequent intervals.                                                                                                                                                                             |
| 7. | Long-term viability                    | In case of any major change in the cloud computing service provider (e.g., acquisition and merger, partnership breakage), the service provided is at the stake.                                                    | Organization should ensure getting their data in case of such major events.                                                                                                                                                                                                                       |

Source: <http://www.infoworld.com/d/security-central/gartner-seven-cloud-computing-security-risks-853> (9 October 2009).

## Course Outcomes (COs)

Upon completion of course the students will be able to

- **analyze cyber offences** and implement effective countermeasures